HDOC DAY-67

Recursive Function to Find the GCD Using Euclid’s Algorithm

Aim

To write a C program that finds the Greatest Common Divisor (GCD) of two positive integers using a recursive function and Euclid’s algorithm.

Algorithm

1. Start.
2. Read two positive integers, a and b.
3. If b is equal to 0, return a.
4. Otherwise, call the function again with b and a % b.
5. Display the returned GCD.
6. Stop.

C Program

```
#include <stdio.h>

int gcd(int a, int b)
{
    if (b == 0)
        return a;
    return gcd(b, a % b);
}

int main(void)
{
    int a, b;

    printf("Enter two positive integers: ");
    scanf("%d %d", &a, &b);

    if (a <= 0 || b <= 0)
    {
        printf("Please enter positive integers only.\\n");
        return 1;
    }

    printf("GCD = %d\\n", gcd(a, b));
    return 0;
}
```

Sample Output

```
Enter two positive integers: 48 18
GCD = 6
```

Explanation

Euclid’s algorithm uses the rule GCD(a, b) = GCD(b, a % b). The recursive calls continue until the second number becomes 0. At that point, the first number is the GCD. For example, GCD(48, 18) becomes GCD(18, 12), then GCD(12, 6), then GCD(6, 0), so the answer is 6.